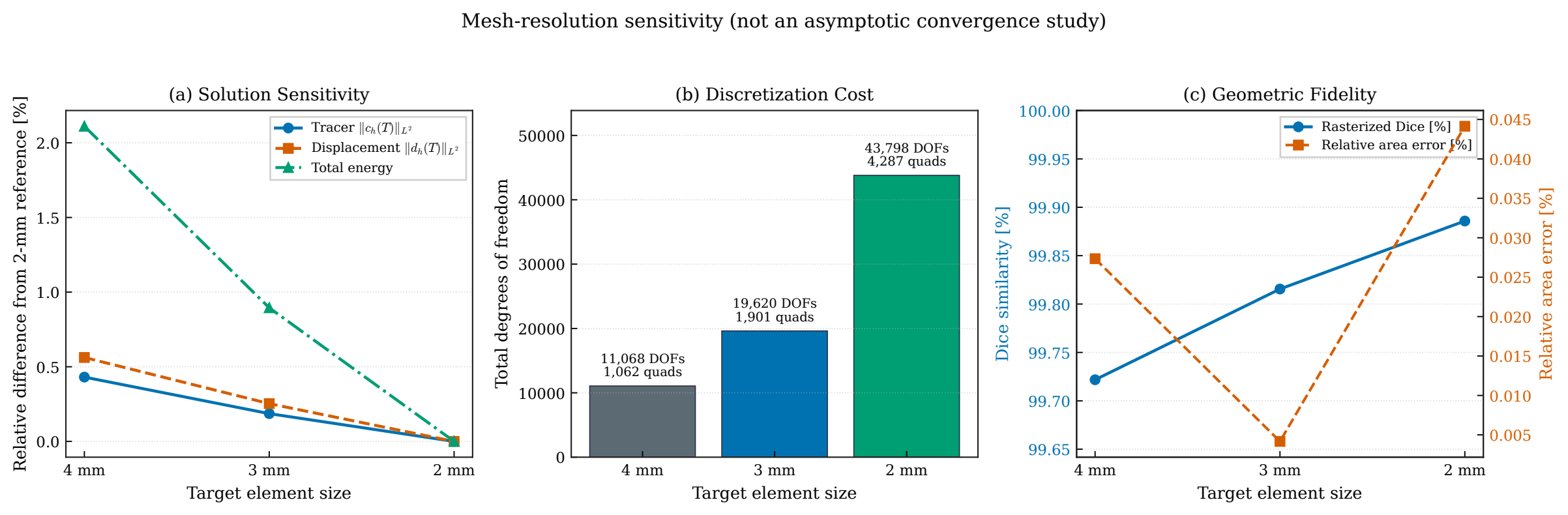
Mesh-resolution sensitivity (not an asymptotic convergence study)
(a) Solution Sensitivity
Relative difference from 2-mm reference [%]
Tracer $\|c_h(T)\|_{L^2}$
Displacement $\|d_h(T)\|_{L^2}$
Total energy
0.0
0.5
1.0
1.5
2.0
4 mm
3 mm
2 mm
Target element size
(b) Discretization Cost
Total degrees of freedom
0
10000
20000
30000
40000
50000
11,068 DOFs
1,062 quads
19,620 DOFs
1,901 quads
43,798 DOFs
4,287 quads
4 mm
3 mm
2 mm
Target element size
(c) Geometric Fidelity
Dice similarity [%]
Relative area error [%]
Rasterized Dice [%]
Relative area error [%]
99.65
99.70
99.75
99.80
99.85
99.90
99.95
100.00
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
4 mm
3 mm
2 mm
Target element size